# SS1 DIGITAL TECHNOLOGIES LESSON NOTES (SECOND TERM)

## WEEK 1: Computer Networks

**Topic:** Computer Networks

**Introduction:**
A computer network is a collection of two or more computers and other hardware devices (like printers and phones) that are interconnected by communication channels (cables or wireless signals) for the purpose of sharing resources and data.

**Subtopic 1.1: Types of Networks (by Geographical Scope)**

1. **PAN (Personal Area Network):**

- **Scope:** Very small, centered around an individual person (within about 10 meters).
- **Purpose:** Connects personal devices for an individual.
- **Example:** Connecting your phone to a Bluetooth earpiece, or a laptop to a wireless mouse.

2. **LAN (Local Area Network):**

- **Scope:** A single, limited geographical area like one building, a school campus, or a home.
- **Purpose:** Sharing resources like printers and files within a single organization.
- **Example:** Your school's computer lab, your home Wi-Fi network.

3. **MAN (Metropolitan Area Network):**

- **Scope:** Spans a city or a large town. It's a larger version of a LAN.
- **Purpose:** Connects multiple LANs across a city.
- **Example:** A university connecting its main campus LAN with its teaching hospital LAN downtown.

4. **WAN (Wide Area Network):**

- **Scope:** A very large geographical area, such as a country, a continent, or the entire globe.
- **Purpose:** Connects multiple LANs and MANs over long distances.
- **Example:** The Internet is the largest WAN. A bank connecting its Lagos HQ LAN to its Abuja branch LAN.

**Subtopic 1.2: Network Architectures (How Devices Behave)**

1. **Client-Server Architecture:**

- **Concept:** A centralized model. One or more powerful computers called **Servers** provide resources (like files or websites), and many **Clients** (your PC, phone) request them.
- **Analogy:** A restaurant. You (the client) order from a waiter, who gets the food from the kitchen (the server).
- **Examples:** A web server, a file server, an email server.

2. **Peer-to-Peer (P2P) Architecture:**

- **Concept:** A decentralized model where every computer (a "peer") is equal. Each device can act as both a client and a server.
- **Analogy:** A potluck dinner. Everyone brings a dish (acts as a server) and everyone eats from other dishes (acts as a client).
- **Examples:** Torrent file-sharing (e.g., BitTorrent), some older home networking setups.

**Subtopic 1.3: Network Topologies (The "Map" of the Network)**

Topology refers to the physical or logical layout of a network.

1. **Bus Topology:** All devices share a single common cable (the "bus").

- **Pro:** Cheap and easy to set up.

- **Con:** If the main cable breaks, the entire network fails.

2. **Star Topology:** All devices connect to a central hub or switch.

- **Pro:** Reliable (if one device's cable breaks, only that device fails; the rest of the network is fine). This is the most common topology used in modern LANs.
- **Con:** If the central hub/switch fails, the entire network fails.

3. **Ring Topology:** All devices are connected in a circle, with data passing from one to the next.

- **Pro:** Orderly data transfer.
- **Con:** If one device or cable breaks, the entire network fails.

4. **Mesh Topology:** Every device is connected to every other device.

- **Pro:** Extremely reliable (many paths for data).
- **Con:** Very expensive and complex to cable. Used in the core of the Internet.

**Subtopic 1.4: Components of a Network**

- **NIC (Network Interface Card):** A piece of hardware inside a computer that allows it to physically connect to the network (via Ethernet cable or Wi-Fi).
- **Switch:** A "smart" central device used in a Star topology. It receives data and sends it only to the specific device it's meant for.
- **Router:** A device that connects different networks together and directs traffic between them. Your home Wi-Fi "router" connects your home LAN to the Internet WAN.
- **Modem (Modulator/Demodulator):** A device that converts the digital signal from your computer into an analog signal to travel over phone or cable lines, and vice-versa.
- **Transmission Media:** The physical "channel" for data.

- **Wired:** Ethernet Cables, Coaxial Cables, Fibre-Optic Cables (fastest, uses light).
- **Wireless:** Radio waves (Wi-Fi, Bluetooth), Microwaves.

**Evaluation Questions:**

1. What is the main difference in scope between a LAN and a WAN?
2. Give a real-world example of a PAN.

3. Explain the difference between a "Client-Server" and a "Peer-to-Peer" network model.
4. What is the most common network topology used today, and why is it more reliable than a Bus topology?
5. What is the specific job of a Router?

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## WEEK 2: Internet Technologies

**Topic:** Internet Technologies

**Introduction:**
The **Internet** (with a capital "I") is the single largest, global WAN. It is a "network of networks," connecting billions of devices. This week, we focus on how the public Internet works.

**Subtopic 2.1: Core Protocols: The "Language" of the Internet**
A protocol is a set of rules that governs communication.

- **TCP/IP (Transmission Control Protocol / Internet Protocol):** This is the foundational suite of protocols for the internet.
  - **IP's Job (The "Address"):** IP is responsible for addressing. Every device on the internet has a unique **IP Address** (e.g., `172.217.14.228`). The IP protocol puts this "address label" on data packets.
  - **TCP's Job (The "Delivery"):** TCP is responsible for reliability. It breaks data into packets, numbers them, ensures they all arrive safely, and reassembles them in the correct order. If a packet is lost, TCP re-sends it.
- **HTTP (HyperText Transfer Protocol):** The protocol used for viewing web pages.
- **HTTPS (HTTP Secure):** A secure, encrypted version of HTTP. The "S" means your communication with the website (e.g., your bank) is private. Always look for the padlock icon.

- **DNS (Domain Name System):** The "phonebook" of the internet. Humans remember domain names (e.g., `google.com`), but computers need IP addresses. A DNS server translates the domain name you type into the correct IP address.

**Subtopic 2.2: Internet Services**

The Internet is the "road." These are the "cars and services" that use the road.

1. **World Wide Web (WWW):** The most popular service. It is a collection of all public websites stored on web servers. (Note: The Internet and the WWW are not the same thing. The WWW runs *on* the Internet).
2. **Email (Electronic Mail):** A service for sending digital messages.
3. **File Transfer Protocol (FTP):** A service for transferring large files.
4. **Voice over IP (VoIP):** A service that lets you make phone calls over the internet (e.g., WhatsApp Calls, Skype).

**Subtopic 2.3: Web Browsers**

- **Definition:** A piece of application software used to request, retrieve, and display web pages.
- **Job:** It sends an HTTP request, receives the files from a server, and renders them into the visual website you see.
- **Examples:** Google Chrome, Mozilla Firefox, Apple Safari, Microsoft Edge.

**Subtopic 2.4: Search Engines**

- **Definition:** A website (a service) that helps you find other websites on the WWW.
- **How they work (The 3 Steps):**

1. **Crawling:** The search engine sends out automated programs called "crawlers" that follow links from one website to the next to discover all public pages.
2. **Indexing:** The crawler sends copies of these pages back, which are analyzed and stored in a massive database called an **Index**.

3. **Ranking:** When you type a search query, the search engine searches its Index and uses a complex algorithm (like Google's PageRank) to decide which pages are the most relevant and lists them in that order.

- **Examples:** Google, Bing, DuckDuckGo.

**Evaluation Questions:**

1. What is the difference between the Internet and the World Wide Web?
2. In the TCP/IP suite, what is the job of IP and what is the job of TCP?
3. What is the purpose of the DNS?
4. Explain the 3 steps a search engine uses to find and list websites.
5. What is the difference between HTTP and HTTPS? Why is HTTPS important?

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## WEEK 3: Email & Online Communication

**Topic:** Email & Online Communication

**Introduction:**
Beyond the web, the most common use of the internet is communication. This ranges from formal (email) to real-time (instant messaging).

**Subtopic 3.1: Email (Electronic Mail)**
Email is an **asynchronous** communication tool, meaning the recipient does not have to be online at the same time as the sender.

- **Anatomy of an Email Address:** `username@domain.com`
  - `username`: The unique identifier for your mailbox.
  - `@`: The "at" symbol, separating the user from the domain.
  - `domain.com`: The domain name of the server that hosts your email (e.g., gmail.com).
- **Email Protocols (The "Post Office" Rules):**

- **SMTP (Simple Mail Transfer Protocol):** The protocol used for **sending** email.
- **POP3 (Post Office Protocol 3):** An older protocol for **receiving** email. It downloads the email from the server to your device and then deletes it from the server. This makes it hard to access emails from multiple devices.
- **IMAP (Internet Message Access Protocol):** A modern protocol for **receiving** email. It syncs your email with the server. The email stays on the server, allowing you to access and manage your email from any device, and everything stays in sync.

**Subtopic 3.2: Email Etiquette (The "Unwritten Rules" of Professionalism)**

1. **Use a Clear, Professional Subject Line:** (e.g., "Question about SS1 Digital Tech Assignment"). Never leave it blank.
2. **Use a Professional Salutation:** (e.g., "Dear Mr. Adebayo,").
3. **Be Concise:** Keep the message clear and to the point.
4. **Do NOT use ALL CAPS:** It is considered "shouting."
5. **Proofread:** Check for spelling and grammar errors before sending.
6. **Reply-All:** Do not use "Reply-All" unless everyone on the list needs to see your reply.

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**Subtopic 3.3: Instant Messaging (IM)**

- **Definition:** A **synchronous** (or near-synchronous) communication tool for real-time text-based chat.
- **Characteristics:** Casual, fast-paced, uses "chats" instead of "messages."
- **Examples:** WhatsApp, Telegram, Facebook Messenger, Slack.

**Subtopic 3.4: Video Conferencing**

- **Definition:** A tool for real-time, visual, and audio communication between two or more people in different locations.
- **Characteristics:** Replicates face-to-face meetings.
- **Examples:** Zoom, Google Meet, Microsoft Teams.

**Evaluation Questions:**

1. What does "asynchronous" mean in the context of email?
2. What is the difference between the SMTP protocol and the IMAP protocol?
3. Why is IMAP generally preferred over POP3 today?
4. List three rules of professional email etiquette.
5. What is the main difference between Instant Messaging and Email?

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## WEEK 4: Digital Storage & Cloud Computing

**Topic:** Digital Storage & Cloud Computing

**Introduction:**
This week, we revisit digital storage in more depth and introduce "Cloud Computing," which combines storage and networking.

**Subtopic 4.1: Review: Types of Digital Storage**

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1. **Magnetic Storage:** Stores data by magnetizing tiny particles on a spinning disk.
   - **Example:** HDD (Hard Disk Drive).
   - **Pro:** Very cheap, large capacity.
   - **Con:** Slow, has moving parts, can be damaged by dropping.
2. **Optical Storage:** Stores data as microscopic "pits" and "lands" on a disk, read by a laser.
   - **Example:** CD-ROM, DVD-ROM.
   - **Pro:** Cheap to manufacture.
   - **Con:** Very slow, small capacity, easily scratched.
3. **Solid-State Storage:** Stores data electronically in flash memory chips.
   - **Example:** SSD (Solid State Drive), USB Flash Drive, SD Card.
   - **Pro:** Extremely fast, no moving parts, durable.
   - **Con:** More expensive than HDDs (but prices are dropping).

### Subtopic 4.2: What is Cloud Computing?

- **Core Idea:** Instead of owning and maintaining your own computer hardware (a "server"), you rent it from another company (a "cloud provider").
- **Definition:** Cloud Computing is the on-demand delivery of IT resources (like storage, processing power, software) over the Internet, with "pay-as-you-go" pricing.
- **Examples of Providers:** Amazon Web Services (AWS), Microsoft Azure, Google Cloud.

### Subtopic 4.3: Cloud Service Models (The "Types" of Cloud)

1. **IaaS (Infrastructure as a Service):**
   - **What you rent:** The basic building blocks—virtual computers (servers), storage, and networking.
   - **Analogy:** You are renting an empty kitchen. You have to bring your own ingredients and pots and pans.
   - **Used by:** System Administrators.
2. **PaaS (Platform as a Service):**
   - **What you rent:** A platform to build and run your own applications. The provider manages the server, OS, and database for you.
   - **Analogy:** You are renting a "pizza-as-a-service" kitchen. The oven and ingredients are provided. You just have to make the pizza.
   - **Used by:** Software Developers.
3. **SaaS (Software as a Service):**
   - **What you rent:** A finished application. You just log in and use it.
   - **Analogy:** You are ordering a finished, hot pizza. You just eat it.
   - **This is the most common model for normal users.** You don't own the software; you just use it.
   - **Examples:** Gmail, Google Docs, Netflix, Zoom.

### Subtopic 4.4: Advantages of Cloud Computing & Data Backup

- **Accessibility:** Access your files and apps from any device, anywhere.
- **Cost:** Cheaper than buying your own server (no upfront cost).
- **Scalability:** You can easily "scale up" (get more power) or "scale down."
- **Data Backup:** The cloud is a perfect solution for keeping an off-site copy of your important files, safe from local disasters like fire or theft.

**Evaluation Questions:**

1. List the three main types of storage and give one example of each.
2. Define "Cloud Computing" in your own words.
3. What is the most common cloud service model that you use every day? Give two examples.
4. Explain the "SaaS" model using an analogy.
5. How does cloud storage help you keep your data safe?

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## WEEK 6: Cybersecurity I - Threats

**Topic:** Cybersecurity I: Threats

**Introduction:**
Cybersecurity is the practice of protecting computer systems, networks, and data from digital attacks, theft, or damage. This week, we focus on the threats.

**Subtopic 6.1: The "CIA Triad" (The Goal of Cybersecurity)**
Cybersecurity aims to protect three things:

1. **Confidentiality:** Keeping data secret (e.g., only you can see your bank balance).
2. **Integrity:** Keeping data accurate (e.g., a hacker cannot change your bank balance).
3. **Availability:** Ensuring data is accessible when you need it (e.g., a hacker cannot shut down your bank's website).

**Subtopic 6.2: Threat 1: Malware (Malicious Software)**

Malware is any software intentionally designed to cause damage.

- **Virus:** A piece of code that attaches itself to a clean program and requires a human to run that program to spread.
- **Worm:** A standalone program that replicates itself and spreads automatically over a network, without human help. It causes fast-moving outbreaks.
- **Trojan (or "Trojan Horse"):** Software that disguises itself as something useful (like a free game). When run, it opens a "backdoor" for attackers.
- **Spyware:** Malware that secretly spies on you. It can record your keystrokes (a Keylogger) to steal passwords, or turn on your webcam.
- **Ransomware:** The most dangerous threat today. It encrypts your files (making them unreadable) and demands a ransom payment to unlock them.

**Subtopic 6.3: Threat 2: Social Engineering (Hacking the Human)**

This is the art of manipulating people to give up confidential information. It's a psychological trick, not a technical hack.

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- **Phishing:** A fraudulent email, text (**Smishing**), or WhatsApp message that pretends to be from a legitimate company (e.g., "Your bank account has been locked! Click here to verify"). The goal is to trick you into clicking a link that installs malware or leads to a fake website to steal your password.
- **Baiting:** Leaving an infected USB drive labeled "Confidential Salaries" on the floor. Curiosity leads someone to plug it in, infecting the computer.

**Subtopic 6.4: Threat 3: Hacking & Direct Attacks**

- **Hacking:** Gaining unauthorized access to a computer system or network.
- **Man-in-the-Middle (MITM) Attack:** An attacker secretly places themselves between you and a website (e.g., on unsecured public Wi-Fi) to read or change your data.
- **Denial of Service (DoS) Attack:** An attacker floods a server with so much junk traffic that it gets overwhelmed and crashes, becoming "unavailable."

**Evaluation Questions:**

1. What are the three components of the "CIA Triad"?
2. What is the main difference between a Virus and a Worm?
3. Explain what a Trojan does.
4. What is Phishing?
5. What is Ransomware and why is it so dangerous?

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## WEEK 8: Cybersecurity II - Defense

**Topic:** Cybersecurity II: Defense

**Introduction:**
Now that we know the threats, we can learn how to defend against them. Cybersecurity defense is about creating layers of protection.

**Subtopic 8.1: Protection 1: Antivirus & Anti-malware**

- **What it is:** A utility program designed to detect, prevent, and remove malware.
- **How it works:**
  - **Signature-Based Detection:** It keeps a database of known malware "fingerprints" and scans files to compare them. (This is why it must be updated daily).
  - **Heuristic-Based Detection:** It looks for suspicious behavior (e.g., "Why is this program trying to encrypt all my files?"). This can catch new, unknown malware.

**Subtopic 8.2: Protection 2: Firewalls**

- **What it is:** A network security device (hardware or software) that acts as a "gatekeeper" for your network. It monitors incoming and outgoing traffic.

- **Analogy:** It is like a bouncer at a club. It checks the "guest list" and only lets approved traffic in or out.

**Subtopic 8.3: Protection 3: Passwords & Authentication**

Authentication is the process of proving you are who you say you are.

1. **Strong Passwords:** A strong password is based on length and complexity. Use at least 12 characters with a mix of uppercase, lowercase, numbers, and symbols.
   A **passphrase** (e.g., `Correct!Horse!Battery!Staple!`) is long, easy to remember, and very hard to crack.
2. **Multi-Factor Authentication (MFA) / 2FA:** This requires more than one "factor" to log in.

- Factor 1: Something you **KNOW** (your password).
- Factor 2: Something you **HAVE** (your phone, which receives a one-time code).
- Even if a hacker steals your password, they cannot log in without your phone. This is the most important way to protect your accounts.

**Subtopic 8.4: Protection 4: Encryption**

- **What it is:** The process of converting plain text (readable data) into scrambled ciphertext (unreadable data) using a "key." The only way to read it is with the correct key.
- **Why it's important:**

- It's the "S" in HTTPS, scrambling your data as it travels over the internet.
- It protects your data if your laptop is stolen (by encrypting the hard drive).
- It's what makes WhatsApp secure (End-to-End Encryption).

**Evaluation Questions:**

1. Explain the two ways an Antivirus program detects malware.
2. What is the main job of a Firewall?
3. What are the two key components of a strong password?
4. What is "2FA" (Two-Factor Authentication)?

5. Why is 2FA so much more secure than just a password?

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## WEEK 9: Data Privacy & Digital Ethics

**Topic:** Data Privacy & Digital Ethics

**Introduction:**
Just because something is possible with technology, it does not mean it is right. Ethics is the study of "what is right and wrong." Data privacy is one of the most important ethical issues today.

**Subtopic 9.1: Data Privacy**

- **Data Privacy** is the right of an individual to control what information is collected about them, and how it is used, stored, and shared.
- **Your "Digital Footprint":** This is the trail of data you leave behind online.
  - **Active Footprint:** Data you intentionally share (e.g., your posts on Instagram).
  - **Passive Footprint:** Data you leave unintentionally (e.g., your IP address, search history, location data from your phone).

**Subtopic 9.2: Data Protection Laws**
Because data is so valuable, governments have created laws to protect citizens.

- **NDPR (Nigeria Data Protection Regulation):** Nigeria's main data privacy law. It gives you the **Right to Consent** (a company must ask your permission to collect your data) and the **Right to be Forgotten** (you can ask a company to delete your data).
- **GDPR (General Data Protection Regulation):** Europe's very strict privacy law, which is the global "gold standard."

**Subtopic 9.3: Digital Ethics: Responsible Use**

- **Intellectual Property & Copyright:** It is unethical to pirate (illegally download) software, movies, or music. This is theft of the creator's work.
- **"Fake News" & Disinformation:** It is unethical to knowingly share false information that could cause panic or harm.

**Subtopic 9.4: Cyberbullying**

- **Definition:** The use of digital technology to repeatedly and intentionally harass, threaten, or humiliate someone.
- **Why it's worse than regular bullying:**

1. **It's 24/7:** The victim cannot escape it, even at home.
2. **It's Permanent:** What goes online can stay online forever.
3. **It's (semi-)Anonymous:** Bullies can hide behind fake profiles.

- **What to do:**

1. **DON'T** respond.
2. **BLOCK** the bully.
3. **SCREENSHOT** the evidence.
4. **REPORT** the bully to the platform, your parents, or a teacher.

**Evaluation Questions:**

1. What is "Data Privacy"?
2. What is the difference between an "Active" and a "Passive" digital footprint?
3. What is the "NDPR"?
4. Define "Cyberbullying" in your own words.
5. List two reasons why cyberbullying can be worse than in-person bullying.

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## WEEK 10: Social Media Technologies

**Topic:** Social Media Technologies

**Introduction:**

Social media platforms are powerful digital technologies that have changed how we socialize, get news, and view the world. They are a key part of "digital citizenship."

**Subtopic 10.1: The Benefits of Social Media**

1. **Connection:** Stay in touch with friends and family.
2. **Information & News:** A fast way to get real-time, breaking news.
3. **Community:** Find and connect with people who share your niche hobbies.
4. **Advocacy & Activism:** A powerful tool for raising awareness.
5. **Opportunity:** A platform for business, art, and personal branding.

**Subtopic 10.2: The Risks & Challenges of Social Media**

1. **Mental Health:** Can lead to anxiety and low self-esteem due to constant social comparison.
2. **Addiction:** These apps are designed to be addictive, using "infinite scroll" and "likes."
3. **"Filter Bubbles" & "Echo Chambers":** The algorithm learns what you like and only shows you content you'll agree with. You end up in a "bubble" where you only hear your own beliefs "echoed" back at you.
4. **Disinformation ("Fake News"):** False information spreads faster on social media than true information.
5. **Privacy:** You "pay" for "free" social media with your data (your clicks, likes, location). This data is sold to advertisers.

**Subtopic 10.3: Responsible Digital Citizenship**

A responsible digital citizen practices:

1. **Critical Thinking:** THINK before you share. (Is this True? Helpful? Inspiring? Necessary? Kind?)
2. **Empathy:** Remembering there is a real human being behind the other screen.

3. **Media Literacy:** Understanding that algorithms create "filter bubbles" and actively seeking out different points of view.
4. **Digital Well-being:** Knowing when to log off and curating your feed to follow positive and educational accounts.

**Evaluation Questions:**

1. List two benefits and two risks of social media.
2. What is a "filter bubble" or "echo chamber"?
3. Explain the business model of most "free" social media companies. (What is the product?)
4. What is "Media Literacy"?
5. List two qualities of a "Responsible Digital Citizen."

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## WEEK 11: Revision

**Topic:** Comprehensive Revision of All Term Topics

**Introduction:**
This week is dedicated to a comprehensive review of all topics covered in the Second Term, in preparation for your exams.

**Revision Checklist:**

- **Week 1 (Networks):**
  - Types: PAN, LAN, MAN, WAN.
  - Architectures: Client-Server vs. Peer-to-Peer.
  - Topologies: Bus, Star, Ring, Mesh (pros/cons).
  - Components: NIC, Switch, Router, Modem.
- **Week 2 (Internet Technologies):**

- TCP/IP: IP (addressing) vs. TCP (reliability).
- DNS (Domain Name System - the phonebook).
- HTTP vs. HTTPS.
- WWW vs. The Internet.
- Browsers vs. Search Engines (Crawling, Indexing, Ranking).

- **Week 3 (Email & Communication):**

- Email protocols: SMTP (send), POP3 (download), IMAP (sync).
- Email Etiquette.
- IM (synchronous) vs. Email (asynchronous).

- **Week 4 (Storage & Cloud):**

- Storage types: Magnetic (HDD), Optical (CD), Solid-State (SSD).
- Cloud Computing: IaaS (rent kitchen), PaaS (rent pizza kitchen), SaaS (order pizza).

- **Week 6 (Cybersecurity I - Threats):**

- CIA Triad (Confidentiality, Integrity, Availability).
- Malware: Virus, Worm, Trojan, Ransomware, Spyware.
- Social Engineering: Phishing, Smishing.

- **Week 8 (Cybersecurity II - Defense):**

- Antivirus (Signatures, Heuristics).
- Firewall (Packet filtering).
- Passwords & 2FA (Know + Have).
- Encryption.

- **Week 9 (Data Privacy & Ethics):**

- Digital Footprint: Active vs. Passive.
- NDPR.
- Cyberbullying (what to do).

- **Week 10 (Social Media):**

- Risks: Filter Bubbles, Echo Chambers, Addiction.
- Benefits: Connection, Information.
- Responsible Digital Citizenship.

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## WEEK 12: Examination

**Topic:** Second Term Examination

This week is for your final examinations on all topics covered in the Second Term. Read each question carefully and do your best.

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## WEEK 13: Marking and Review of Scripts

**Topic:** Marking and Review of Scripts

Your teacher will review the exam with you. Pay close attention to the corrections to learn from any mistakes and ensure you understand all the concepts from the term. This is a valuable part of your learning process.

**Good luck!**

*This response is AI-generated, for reference only.*